



The Higher Institute of
Engineering
El Shorouk City

Program: Computer and Control Eng. (CCE)

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Assignment No. 1 (Open Book)

Course Code: CCE 463

Level: 4th year (Reg. 2019)

Course Name: Networks (2)

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هندسة الاتصالات والحاسبات
Communications and Computer Engineering Dept.

Communications and Computer
Engineering Dept.

Assignment No. (1)

Students must address the following topics and fulfil all stated requirements for each topic:

- **Data Center:** Compare traditional, modern, and smart data centers in terms of definition; structural architecture (design model only); importance and use cases; core components; internal network technologies (routing, switching, and automation); and cloud integration (external connectivity and workload integration). Additionally, identify the most prominent vendors supplying data center solutions, discuss the expected cost ranges, and highlight the leading employment companies operating in this field.
- **Cloud Computing:** Define cloud computing and explain its importance; computing virtualization; storage architecture; key virtualization features; the most prominent vendors providing cloud computing services; expected cost ranges; and the leading employment companies operating in this field.
- **Software-Defined Networking (SDN) and Network Function Virtualization (NFV):** Define SDN and NFV and explain their architecture, benefits, use cases, challenges, and key design considerations.
- **Protocol Overview:** Provide an overview of the following protocols, explaining the function and practical use of each: Cisco Discovery Protocol (CDP), Link Layer Discovery Protocol (LLDP), Internet Protocol Service Level Agreement (IP SLA), Network Time Protocol (NTP), System Logging Protocol (Syslog), Simple Network Management Protocol (SNMP), and Software-Defined Networking (SDN).

Essential Notes (Must Be Read Carefully Before Preparing the Report):

- Scientific references and reputable websites may be used; however, all references used in preparing the assignment must be properly cited. At a minimum, students are required to consult the Egyptian Knowledge Bank (EKB) and IEEE Xplore Digital Library.

- Companies relevant to the assigned topics must be identified, along with indicative pricing ranges related to cloud computing services and data center solutions.
- The assignment must be submitted in one of the following formats: DOCX or PPT. In addition, a recorded voice explanation (no more than 3 minutes) must accompany the submission.
- Students must pay careful attention to the organization, formatting, and overall presentation quality of the assignment.
- Visual elements (figures, diagrams, and images) should be incorporated into the report where appropriate.
- The completed file must be uploaded to the designated submission area on the E-Learning Moodle platform.
- Moodle accepts only DOCX or PPT files. No other file formats will be accepted.
- The student's name, section number, and course name must appear on the first page of the file, as illustrated on the following page.
- The file must be named using the following format:
Sec. (Section Number) – Student Full Name
Example: Sec. (1) – Mohamed Abdelhamed Abdelhamed

Important Notice:

- Failure to comply with the above instructions will result in a deduction of marks.
- Students are expected to complete the assignment independently, relying on educational resources available online. Any attempt to copy from colleagues is strictly prohibited.
- If identical or substantially similar assignment reports are detected, the students involved will receive a zero grade.

Best Regards,
Dr. Mohamed A. Abdelhamed

Total Marks
(10)



El Shorouk Academy

Higher Institute of Engineering

Communications and Computer Engineering Department



Assignment No. (1)

Report on

Data Center, Cloud Computing, Software-defined Networks (SDN) and Network Function Virtualization (NFV), and Network Protocols

One of the requirements for obtaining an oral/practical degree for a
course

Networks (2), CCE 463

Fourth year

Computer and Control Eng. (CCE) Program

Regulation (2019)

Second Semester, 2025/2026

By

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Report on Data Centers, Cloud Computing, SDN/NFV, and Network Protocols

1. Data Centers

In modern information technology environments, data centers play a fundamental role in supporting digital services.

A data center can be defined as a physical facility that hosts computing infrastructure such as servers, storage systems, and networking equipment. Organizations rely on data centers to process, store, and distribute large amounts of data that support applications, enterprise systems, and online services.

Types of Data Centers

Traditional Data Center

Traditional data centers are usually owned and managed by organizations themselves. They rely mainly on dedicated hardware installed on-premises. These environments typically require manual configuration and maintenance.

Most traditional data centers follow a three-tier architecture consisting of the core layer, aggregation layer, and access layer.

Such data centers are commonly used for internal enterprise systems, banking infrastructure, and government information processing where organizations prefer to maintain direct control over their hardware resources.

Modern Data Center

Modern data centers are designed to support virtualization, automation, and scalable infrastructure.

One of the most common architectures used in modern designs is the spine-leaf model, which improves bandwidth distribution and reduces network latency.

Unlike traditional facilities, modern data centers rely heavily on virtualized servers and automated management tools.

This makes them more flexible and easier to scale when demand increases. These

environments are widely used for large enterprise applications, high-performance computing, and cloud service platforms.

Smart Data Center

Smart data centers represent the next step in data center evolution. These facilities integrate advanced technologies such as Artificial Intelligence (AI), Internet of Things (IoT), and predictive analytics to improve performance and operational efficiency.

For example, AI-based monitoring systems can automatically detect potential failures, while intelligent cooling systems can reduce energy consumption. Smart data centers are typically used by hyperscale cloud providers and organizations running large data analytics or AI workloads.

Core Components of Data Centers

A typical data center includes several essential components:

Servers – responsible for running applications and processing data.

Storage Systems – technologies such as Network Attached Storage (NAS), Storage Area Networks (SAN), and distributed storage systems are used to store large volumes of information.

Networking Equipment – switches, routers, and load balancers ensure communication between servers and external networks.

Power Systems – uninterrupted power supplies (UPS) and backup generators guarantee continuous operation.

Cooling Systems – cooling infrastructure prevents overheating and maintains stable hardware performance.

Internal Network Technologies

Data centers rely on multiple networking technologies to ensure reliable communication.

Routing protocols such as OSPF and BGP control how packets move between networks.

Switching technologies allow servers to communicate efficiently within the data center network.

Automation tools are increasingly used to simplify configuration and deployment. Techniques such as Infrastructure as Code (IaC) allow administrators to manage infrastructure using software scripts.

Cloud Integration

Many modern organizations integrate their data centers with cloud platforms. Hybrid cloud models combine local infrastructure with public cloud services, while multi-cloud strategies use several providers to increase reliability and flexibility.

Major cloud platforms include Amazon Web Services (AWS), Microsoft Azure, and Google Cloud Platform.

Major Vendors

Leading companies that provide data center technologies include Cisco, Dell Technologies, Hewlett Packard Enterprise (HPE), and IBM.

Cost Range

The cost of building a data center depends largely on its size and capacity. Small enterprise data centers may cost between \$100,000 and \$1 million, while medium-scale facilities can reach \$10 million. Hyperscale data centers operated by major cloud companies may cost hundreds of millions or even billions of dollars.

Major Employment Companies

Global technology companies that operate large data center infrastructures include Amazon, Microsoft, Google, and Oracle. These companies require professionals specialized in networking, cloud computing, cybersecurity, and infrastructure management.

2. Cloud Computing

Cloud computing refers to the delivery of computing services through the internet. These services include servers, storage, databases, networking, and software applications. Instead of owning physical infrastructure, organizations can access computing resources on demand.

Importance of Cloud Computing

Cloud computing has become essential for modern businesses. It provides several advantages such as cost efficiency, scalability, global accessibility, and faster deployment of applications.

Computing Virtualization

Virtualization is one of the core technologies behind cloud computing. It allows multiple virtual machines to run on a single physical server using software known as a hypervisor.

Common forms of virtualization include:

Server virtualization

Network virtualization

Storage virtualization

Storage Architecture

Cloud storage systems are designed to provide flexible and reliable data storage. Three common storage models are widely used:

Object Storage – typically used for storing large volumes of unstructured data.

Block Storage – commonly used in databases and applications that require high performance.

File Storage – provides shared file systems for users and applications.

Key Virtualization Features

Some important features of virtualization technology include resource pooling, isolation between virtual machines, rapid scalability, and fault tolerance.

Major Cloud Vendors

The most well-known cloud service providers include:

Amazon Web Services (AWS)

Microsoft Azure

Google Cloud Platform

IBM Cloud

Cloud Pricing

Cloud services generally follow a pay-as-you-go model.

For example, a virtual machine instance may cost between \$5 and \$50 per month depending on its specifications.

Storage services may cost approximately \$0.02 to \$0.10 per gigabyte.

Large enterprise deployments may cost hundreds or thousands of dollars monthly.

3. Software Defined Networking (SDN) and Network Function Virtualization (NFV)

Software-Defined Networking (SDN) is a networking architecture that separates the control plane from the data forwarding plane. This separation allows network administrators to manage and control the entire network from a centralized controller.

SDN Architecture

SDN architecture is typically divided into three layers:

Application Layer – contains network applications and services.

Control Layer – includes the SDN controller that manages network behavior.

Infrastructure Layer – consists of physical switches and routers responsible for forwarding traffic.

Benefits of SDN

SDN offers several benefits such as centralized management, improved automation, better scalability, and simplified network configuration.

Use Cases

SDN is widely used in cloud data centers, enterprise networks, and environments that require flexible network management.

Challenges

Despite its advantages, SDN also presents challenges including security concerns, complex deployment processes, and compatibility issues with legacy hardware.

Network Function Virtualization (NFV)

NFV focuses on transforming traditional network hardware functions into software-based services.

Instead of relying on dedicated hardware devices, network services can run on standard servers.

Examples of virtualized network functions include virtual firewalls, virtual routers, and virtual load balancers.

Benefits of NFV

NFV reduces hardware costs, accelerates service deployment, and provides more flexible network management.

4. Protocol Overview

Cisco Discovery Protocol (CDP)

CDP is a proprietary protocol developed by Cisco. It allows Cisco devices to discover information about directly connected neighboring devices, such as device type and IP address.

Link Layer Discovery Protocol (LLDP)

LLDP is a vendor-neutral discovery protocol used by network devices to advertise identity and capabilities to neighboring devices.

IP Service Level Agreement (IP SLA)

IP SLA is a technology used to measure network performance metrics including latency, jitter, and packet loss.

Network Time Protocol (NTP)

NTP is responsible for synchronizing system clocks across network devices to ensure accurate timing.

Syslog

Syslog is a standard protocol used to collect, store, and manage log messages generated by network devices and servers.

Simple Network Management Protocol (SNMP)

SNMP is widely used for monitoring and managing network devices. It allows administrators to collect performance data and detect network issues.

Software Defined Networking (SDN)

SDN is also considered a modern networking approach that allows programmable control over network infrastructure.

References

Kreutz, D., Ramos, F. M., Veríssimo, P., Rothenberg, C., Azodolmolky, S., & Uhlig, S. *Software-Defined Networking: A Comprehensive Survey*. Proceedings of the IEEE, 2015.

Singh, S., & Jha, R. *A Survey on Software Defined Networking: Architecture for Next Generation Network*. IEEE / arXiv, 2020.

Kaljic, E., Maric, A., Njemcevic, P., & Hadzialic, M. *A Survey on Data Plane Flexibility and Programmability in Software-Defined Networking*. IEEE / arXiv, 2019.